

Abdul Wahid

Lab Report

Cyber Counterintelligence – Honeypot Threat Collection (KFSensor)

Executive Summary

Overview

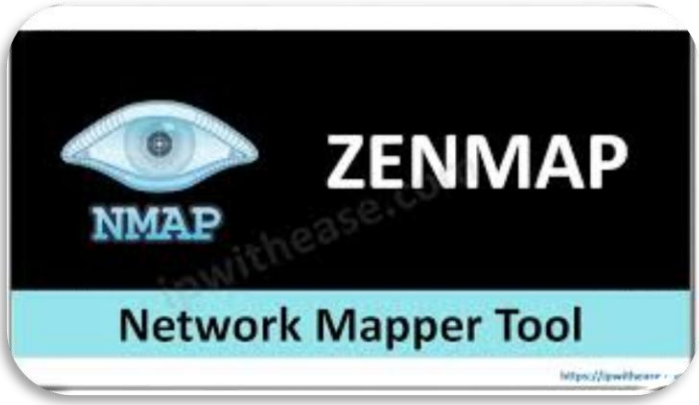
This lab demonstrates the use of KFSensor — a Windows-based honeypot — as a Cyber Counterintelligence (CCI) data collection tool. By deploying a honeypot on a Windows 10 virtual machine and scanning it from Kali Linux using Zenmap (Nmap GUI), the lab captures real-time intrusion attempts and open-port data. This information supports threat analysis and network hardening decisions.

Key Findings

- KFSensor successfully detected and logged all port scan activity initiated by the Kali Linux attacker machine.
- Multiple open TCP ports were identified, providing potential entry points for adversaries.
- The honeypot flagged 'Recent Activity' on services such as FTP, SSH, Telnet, SMTP, and HTTPS.
- CCI tools like honeypots provide actionable intelligence for blocking threats and strengthening network security posture.

Outcome

The analyst gains hands-on experience collecting threat data via a honeypot and learns how to respond — closing unnecessary open ports to reduce attack surface.





Lab Information

Module	Module 04 — Data Collection and Processing
Topic	Data Collection through Cyber Counterintelligence (CCI) Techniques
Primary Tool	KFSensor (Honeypot)
Secondary Tool	Zenmap / Nmap

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Overview of Cyber Counterintelligence (CCI)

CCI is a defensive technique where information about adversaries is gathered to analyze their capabilities and provide necessary protection. A key CCI method is the honeypot.

What is a Honeypot?

A honeypot is a decoy computer system designed to attract and trap unauthorized users attempting to penetrate a network. It:

- Has no legitimate production value — all traffic to it is suspicious
- Logs port access attempts and attacker keystrokes
- Acts as an early warning system for more coordinated attacks
- Provides intelligence on attacker resources, tools, and penetration methods

Lab Environment

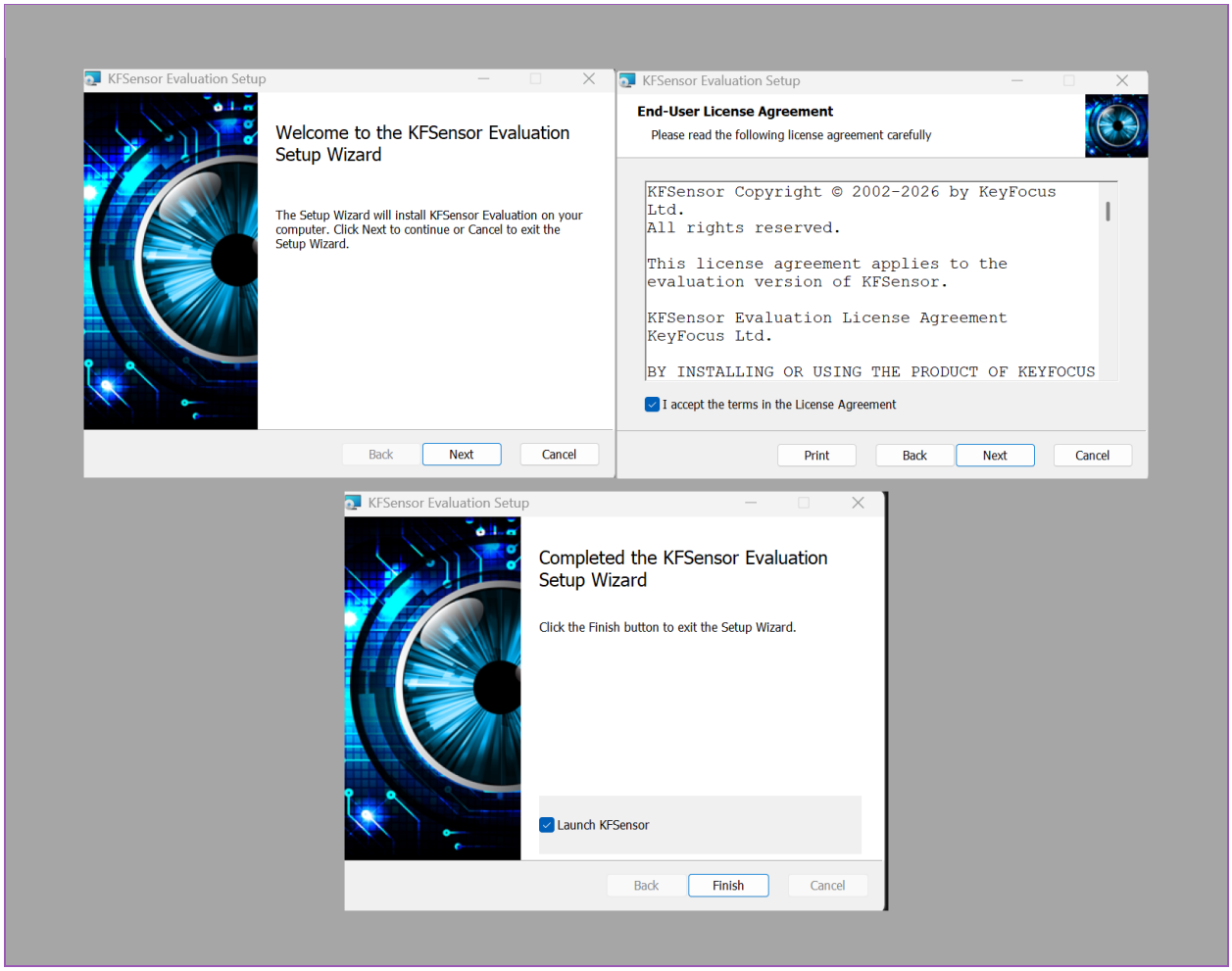
The following components are required to complete this lab:

Windows 10 VM	Target machine where KFSensor honeypot is installed (Username: Analyst | Password: Pa$$w0rd)
Kali Linux VM	Attacker machine used to run Zenmap port scans (Username: root | Password: toor)
KFSensor	Located at D:\CTIA-Tools\CTIA Module 04 Data Collection and Processing\KFSensor
Zenmap	Available via Applications > 01 - Information Gathering > zenmap on Kali Linux
Web Browser	Required for downloading tools or latest versions
WinPcap	Must be installed on Windows 10 before running KFSensor (download from winpcap.org)
Email Account	A dummy email account for KFSensor alert configuration (do not use real credentials)

Task 1: Installing KFSensor on Windows 10

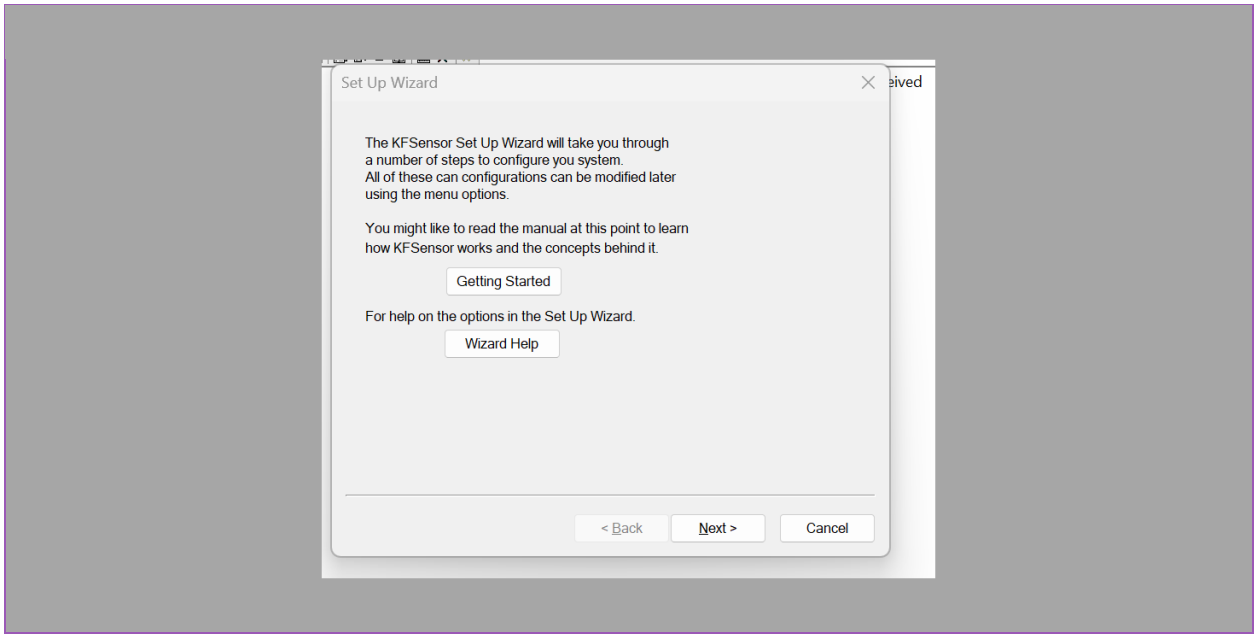
1. Power on the **Windows 10** virtual machine.
2. Log in using credentials — Username: **Analyst** | Password: **Pa$$w0rd**.
3. Navigate to **D:\CTIA-Tools\CTIA Module 04 Data Collection and Processing\KFSensor**, double-click **kfsens40.msi**, and follow the wizard steps to install KFSensor.

NOTE	Ensure WinPcap is installed before running KFSensor. Visit https://www.winpcap.org to download and install it.



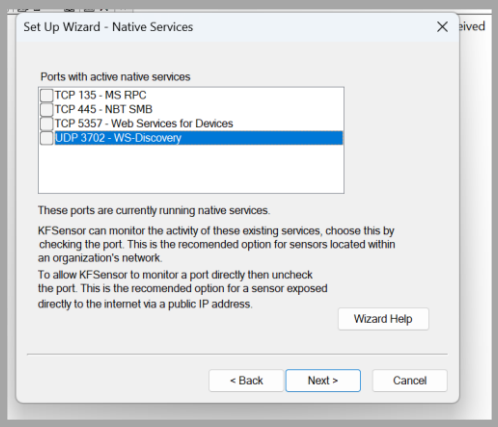
Task 2: Configuring KFSensor via Setup Wizard

4. After installation, KFSensor launches automatically. If not, open it via **Start > Apps > KFSensor > Run as administrator**. The **Set Up Wizard** will appear — click **Next**.



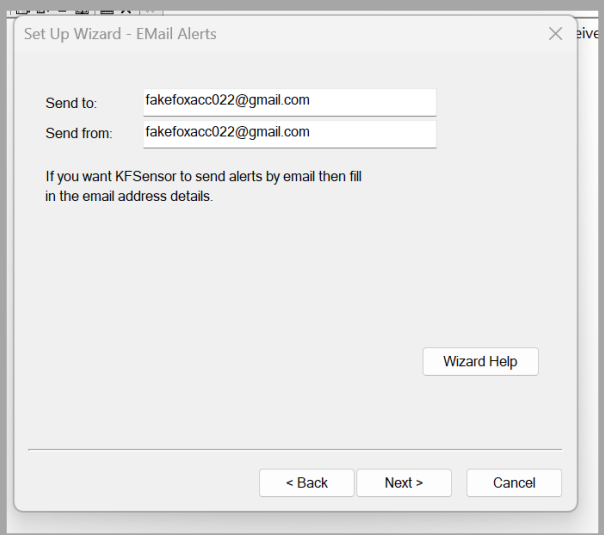
5. In the **Set Up Wizard – Native Services** window, **uncheck all listed ports** (TCP 135, TCP 445, TCP 5357, UDP 3702) to allow KFSensor to monitor them directly. Click **Next**.

WHY?	Unchecking ports allows KFSensor to take full control and monitor those ports directly — the recommended option for sensors exposed to the internet.

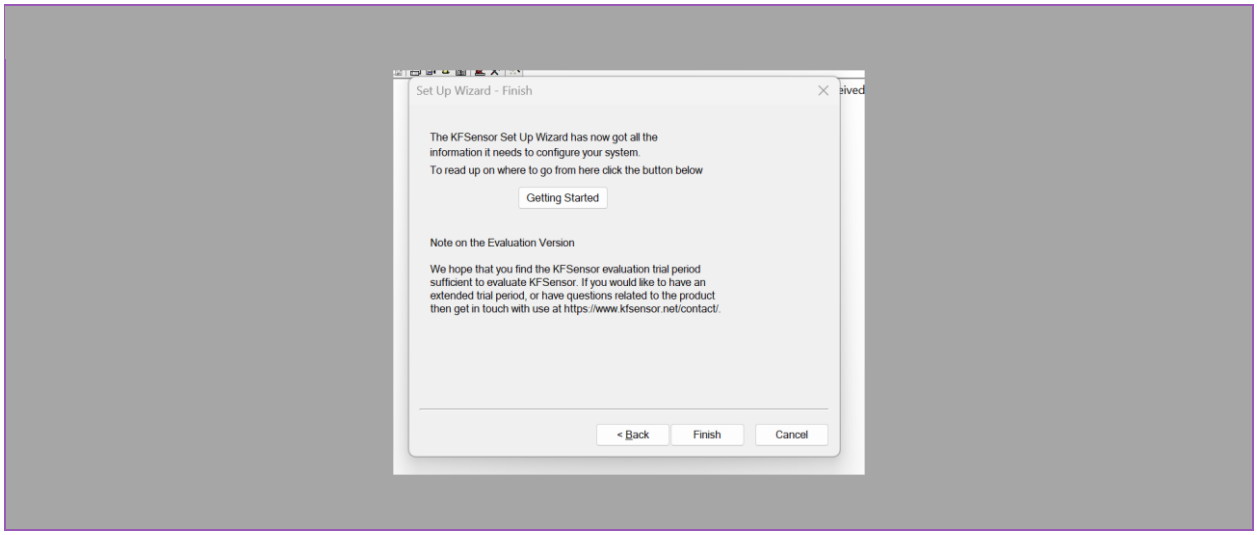


6. In the **Set Up Wizard – Email Alerts** screen, enter a dummy email address in both **Send To** and **Send From** fields. Click **Next**.

NOTE	Use a non-real/dummy email for this lab exercise. Do not use your personal credentials.

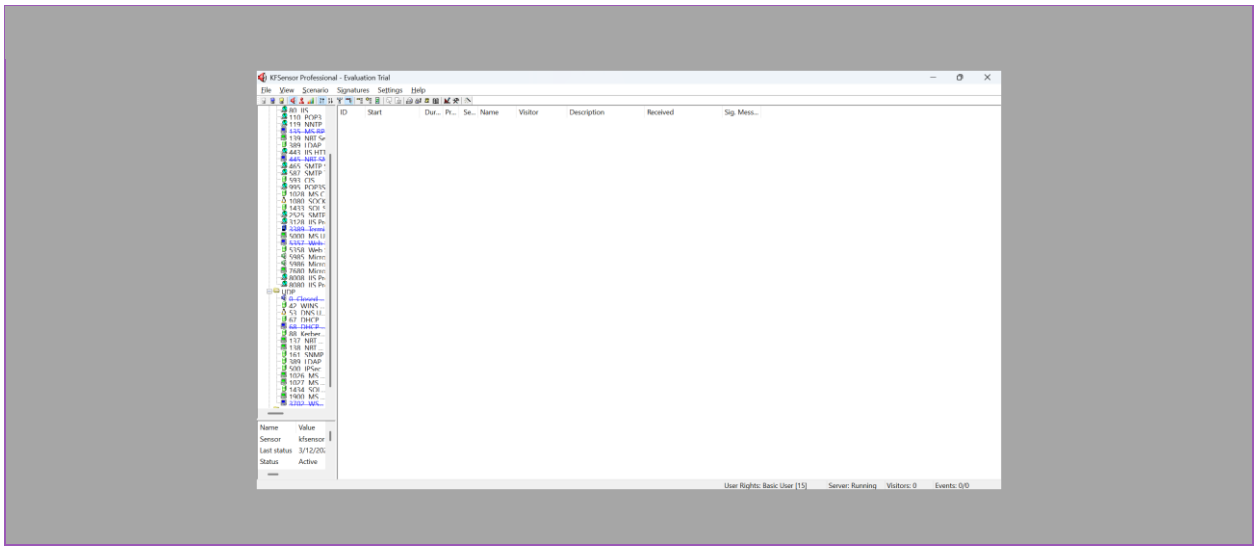


7. Click **Finish** to complete the KFSensor setup wizard.



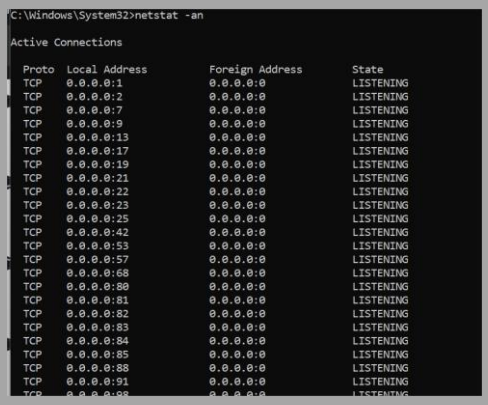
Task 3: Reviewing the KFSensor Main Window

8. The **KFSensor Professional – Evaluation Trial** main window opens. It displays ID, Protocols, Visitor data, and Sensor Port information. Ports currently in use are indicated with blue cross lines in the left panel.



Task 4: Checking Active Listening Ports via Command Prompt

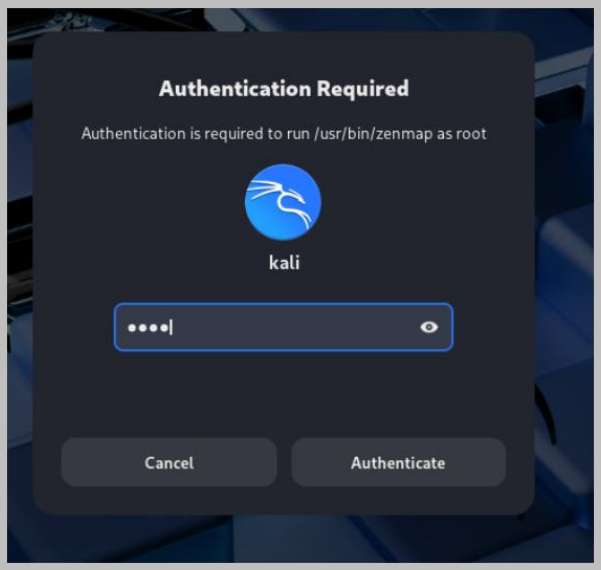
9. Open **Command Prompt** as Administrator from the Apps list.
10. Type the command: **netstat -an** and press Enter.
11. A list of all active **listening ports** is displayed, confirming KFSensor is actively monitoring these ports.



KEEP RUNNING	Leave KFSensor running in the background. Do not close it before proceeding to the next task.

Task 5: Running a Port Scan from Kali Linux (Zenmap)

12. Switch to the **Kali Linux** virtual machine and log in with credentials: root / toor.
13. Navigate to **Applications > 01 - Information Gathering > zenmap** to launch Zenmap.

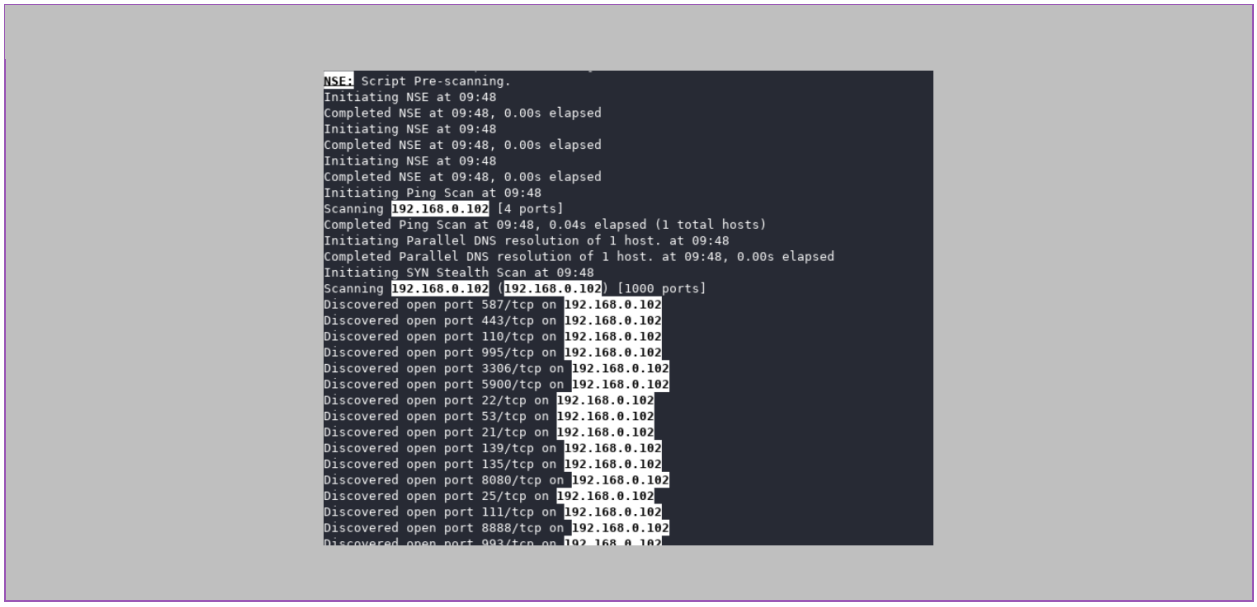


14. In the Zenmap **Target** field, enter the IP address of the Windows 10 machine (e.g., **192.168.0.102**). Set Profile to **Intense Scan** and click **Scan**.

NOTE If authentication is required on Kali, enter the kali user password when prompted by the authentication dialog.

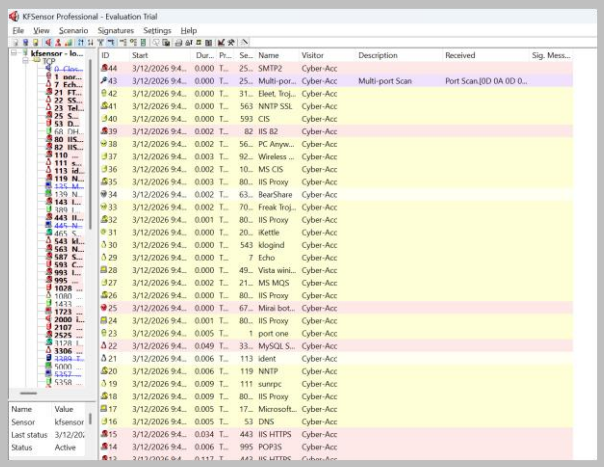


15. Zenmap begins scanning and displays open ports in real time in the Nmap Output tab.
16. The scan results reveal multiple open TCP ports, which are potential entry points for attackers.



Task 6: Observing KFSensor Detection Results

17. Switch back to the **Windows 10** virtual machine.
18. Observe the KFSensor main window. Ports flagged with "**Recent Activity**" (in red) confirm that KFSensor has detected and logged the scan from the Kali Linux machine.



OBSERVATION	KFSensor logged activity across services including FTP (21), SSH (22), Telnet (23), SMTP (25), DNS (53), HTTP (80), HTTPS (443), MySQL (3306), VNC (5900), and more — all flagged as accessed by the Kali attacker machine.

Analysis

What Did We Observe?

KFSensor successfully operated as a honeypot, logging every connection attempt made by the Zenmap scanner. Each open port discovered in the Nmap scan was simultaneously reflected in KFSensor's activity log, demonstrating the honeypot's effectiveness.

Security Implications

- Open ports represent attack vectors — any unnecessary open port should be closed immediately.
- Services like FTP (21), Telnet (23), and VNC (5900) transmit data in plaintext and are particularly dangerous if exposed.
- The attacker machine (Kali) was identified by its IP in the Visitor column — enabling targeted blocking via firewall rules.
- CCI tools like KFSensor provide early warning of reconnaissance activity before a full attack occurs.

Recommended Security Actions

- Close all unnecessary open ports using Windows Firewall or network-level ACLs.
- Disable unused services (e.g., Telnet, FTP) and replace with secure alternatives (SSH, SFTP).
- Monitor honeypot logs regularly and integrate with a SIEM for automated alerting.
- Use the honeypot data to update threat intelligence feeds and firewall rule sets.

Conclusion

This lab successfully demonstrated how KFSensor, a honeypot-based CCI tool, can be deployed to collect threat intelligence data. By simulating an adversarial network scan from Kali Linux and monitoring the results in KFSensor, analysts can identify exposed services, track attacker behavior, and take proactive steps to secure the network.

The integration of CCI techniques into a threat intelligence workflow enables organizations to shift from reactive defense to proactive identification and mitigation of threats — a critical capability for any modern security operations team.